MEME15203 Statistical Inference Marking Guide

Assignment 1

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Due by:			

Q1. Let X_1 and X_2 be a random sample of size $n = 2$ from a continuous distribution with pdf of the form $f(x) = 5x^4$ if $0 < x < 1$ and zero otherwise.

(a) Find the joint pdf of $Y_1 = \min(X_1, X_2)$ and $Y_2 = \max(X_1, X_2)$.

(b) Find the pdf of the sample range $R = Y_2 - Y_1$.

(10 marks)

Ans.

$f_{Y_1,Y_2}(y_1, y_2) = 2!f(y_1)f(y_2) = 2!(5y_1^4)(5y_2^4) = 50y_1^4y_2^4, 0 < y_1 < y_2 < 1$

Making the tranformation $R = Y_2 - Y_1$, $S = Y_1$, yields the inverse tranformation $y_1 = s$, $y_2 = r + s$, and $|J| = 1$. Thus the joint pdf of R and S is

$f_{R,S}(r, s) = f_{Y_1,Y_2}(s, s + t)|J| = 50s(r + s), 0 < s < 1 - r, 0 < r < 1$

$$\begin{aligned} f_R(r) &= \int_0^{1-r} 50s^4(r+s)^4 ds \\ &= \int_0^{1-r} 50s^4(r^4 + 4rs + s^4)ds \\ &= 50[\tfrac{r^4s^3}{3} + \tfrac{rs^4}{2} + \tfrac{s^5}{5}]_0^{1-r} \\ &= 50[\tfrac{r^4(1-r)^3}{3} + \tfrac{r(1-r)^4}{2} + \tfrac{(1-r)^5}{5}] \end{aligned}$$

Q2. Let X_1 and X_2 be independent random variables with $X_1 \sim GAM(\alpha_1 = a, \theta = 2)$ and $X_2 \sim GAM(\alpha_2 = b, \theta = 2)$, show that $U = \frac{X_1}{X_1+X_2}$ follow a Beta distribution. Suppose $Y_i \sim GAM(\alpha = 10, \theta = 2)$, using the result above, find the distribution of $V = \frac{Y_1}{\sum_{i=1}^{16} Y_i}$.

(10 marks)

Ans.

$$f_{X_1,X_2}(x_1, x_2) = \frac{1}{\Gamma(a)\theta^a} x_1^{a-1} e^{-\frac{x_1}{\theta}} \frac{1}{\Gamma(b)\theta^b} x_2^{b-1} e^{-\frac{x_2}{\theta}}, -\infty < x_1 < \infty < x_2 < \infty.$$

Let $u = x_1/(x_1 + x_2)$, $v = x_1 + x_2$, then $x_1 = uv$, $x_2 = v - uv$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v$$

The set $(x_1 > 0, x_2 > 0)$ is mapped to the set $(0 < u < 1, v > 0)$

$f_{U,V}(u,v)$
$= f_{X_1,X_2}(uv, v-uv)|J|$
$= \dfrac{1}{\Gamma(a)\theta^a}(uv)^{a-1}e^{-\frac{uv}{\theta}}\dfrac{1}{\Gamma(b)\theta^b}(v-uv)^{b-1}e^{-\frac{u-uv}{\theta}}|v|$
$= \frac{1}{\Gamma(a)\Gamma(b)\theta^{a+b}}v^{a+b-1}e^{-\frac{v}{\theta}}u^{a-1}(1-u)^{b-1}, 0<u<1, v>0$

$f_U(u)$
$= \int_0^\infty \frac{1}{\Gamma(a)\Gamma(b)}v^{a+b-1}e^{-\frac{v}{\theta}}u^{a-1}(1-u)^{b-1}dv$
$= \frac{1}{\Gamma(a)\Gamma(b)\theta^{a+b}}[\Gamma(a+b)\theta^{a+b}]u^{a-1}(1-u)^{b-1}$
$= \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}u^{a-1}(1-u)^{b-1}, & 0<u,1 \\ 0, & \text{otherwise} \end{cases}$
Hence, $U = X_1/(X_1+X_2)$ is $Beta(a,b)$
$V = \frac{Y_1}{\sum_{i=1}^{16}Y_i} = \frac{Y_1}{Y_1+\sum_{i=2}^{16}Y_i}$
Here, $Y_1 \sim Gamma(\alpha = 10, 2)$ and $\sum_{i=2}^{16}Y_i \sim Gamma(150,2)$, thus, $V \sim Beta(a=10, b=150)$

Q3. The waiting time X until delivery of a new component for an industrial operation is uniformly distributed over the interval from 1 to 9 days. The cost of this delay is given by $U = 5X^2 + 8$. Find the probability density function for U using distribution method.

(10 marks)

Ans.

$X \sim U(1,9)$
$f_X(x) = \frac{1}{9-1}, 1<x<9$
$F_X(x) = \int_1^x \frac{1}{9-1}dt = \frac{x-1}{9-1} = \frac{1}{8}(x-1)$
$F_U(u) = P(U \le u) = P(5X^2+8 \le u) = P(X \le \sqrt{\frac{u-8}{5}}) = \frac{1}{8}\left(\sqrt{\frac{u-8}{5}} - 1\right)$
$f_U(u) = F_U'(u) = \frac{1}{8}\left(\frac{1}{2}\sqrt{\frac{5}{u-8}}\right)\frac{1}{5} = \frac{1}{80}\left(\sqrt{\frac{5}{u-8}}\right), 13<u<413$, zero otherwise.

Q4. In each of the following, random variable X and Y are independent. Use moment generating function methodto find the probability density functions (pdfs)for $V = X+Y$.

(a)$f(x) = \binom{15}{x}0.8^x 0.2^{15-x}, x = 0,1,\ldots,15$
$f(y) = \binom{13}{y}0.8^y 0.2^{13-y}, y = 0,1,\ldots,13$

(b)$X \sim POI(\lambda = 1.6)$
$Y \sim POI(\lambda = 2.7)$

(c)$X \sim N(\mu = 15, \sigma^2 = 5^2)$
$Y \sim N(\mu = 6, \sigma^2 = 7^2)$

(d)$f(x) = \frac{1}{\Gamma(3)6^3} x^2 e^{-\frac{x}{6}}, x > 0$
$f(y) = \frac{1}{\Gamma(3)6^3} y^2 e^{-\frac{y}{6}}, y > 0$

(12 marks)

Ans.

(a)$X \sim Bin(n = 15, p = 0.8)$
$Y \sim Bin(n = 13, p = 0.8)$
$M_X(t) = (0.2 + 0.8e^t)^{15}$
$M_Y(t) = (0.2 + 0.8e^t)^{13}$
$M_{X+Y} = M_X(t)M_Y(t) = (0.2+0.8e^t)^{15}(0.2+0.8e^t)^{13} = (0.2+0.8e^t)^{15+13} = (0.2 + 0.8e^t)^{28}$
Thus, $V + X + Y \sim Bin(n = 28, p = 0.8)$
$\therefore \boxed{f(v) = \binom{28}{v} 0.8^v 0.2^{28-v}, v = 0, 1, \ldots, 28}$

(b)$M_X(t) = e^{1.6(e^t-1)}$
$M_Y(t) = e^{2.7(e^t-1)}$
$M_{X+Y} = M_X(t)M_Y(t) = e^{1.6(e^t-1)}e^{2.7(e^t-1)} = e^{(1.6+2.7)(e^t-1)} = e^{4.3(e^t-1)}$
Thus, $V + X + Y \sim POI(\lambda = 4.3)$
$\therefore \boxed{f(v) = \frac{e^{-4.3}4.3^v}{v!}, v = 0, 1, \ldots}$

(c)$M_X(t) = e^{15t+\frac{1}{2}(5^2)t^2}$
$M_Y(t) = e^{6t+\frac{1}{2}(5^2)t^2}$
$M_{X+Y} = M_X(t)M_Y(t) = e^{15t+\frac{1}{2}(5^2)t^2}e^{6t+\frac{1}{2}(7^2)t^2} = e^{(15+6)t+\frac{1}{2}(5^2+7^2)t^2} = e^{21t+\frac{1}{2}(74)t^2}$
Thus, $V = X + Y \sim N(\mu = 21, \sigma^2 = 74)$
$\therefore \boxed{f(v) = \frac{1}{\sqrt{2\pi(74)}} e^{-\frac{1}{148}(v-21)^2}, -\infty < x < \infty}$

(d)$M_X(t) = (1 - 6t)^{-3}$
$M_Y(t) = (1 - 6t)^{-3}$
$M_{X+Y} = M_X(t)M_Y(t) = (1 - 6t)^{-3}(1 - 6t)^{-3} = (1 - 6t)^{-(3+3)} = (1 - 6t)^{-6}$
Thus. $V = X + Y \sim GAM(\alpha = 6, \theta = 6)$
$\therefore \boxed{f(v)) = \frac{1}{\Gamma(6)6^6} x^5 e^{-\frac{v}{6}}, v > 0}$

Q5. Let Y_n denote the maximum of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $W_n = n[1 - F(Y_n)]$.

(10 marks)

Ans.

$g_n(y_n) = n[F(y_n)]^{n-1} f(y_n), -\infty < y_n < \infty$
$w = n[1 - F_(y_n)]$ and $\frac{dy_n}{dw} = \frac{1}{nf(y_n)}$
$h(w) = [1 - w/n]^{n-1}, 0 < w < n$
$\lim_{n\to\infty} H_n(w) = \lim_{n\to\infty} \int_0^w [1 - u/n]^{n-1} du = \int_0^w e^{-u} du = 1 - e^{-w}$
which is cdf of EXP(1).

Q6. Consider a random sample from a Exponential distribution, $X_i \sim Exp(\theta)$. Find the asymtotic normal distribution of $Y_n = [\ln(\bar{X}_n)]^9$.

(8 marks)

Ans.

$E(\bar{X}_n) = \theta$, $V(\bar{X}_n) = \frac{1}{n}V(X) = \frac{\theta^2}{n}$
By CLT, $\bar{X}_n \sim N\left(\theta, \frac{\theta^2}{n}\right)$
$g(\theta) = (\ln\theta)^9$, $g'(\theta) = \frac{9}{\theta}(\ln\theta)^8$, $[g'(\theta)]^2 = \frac{81}{\theta^2}(\ln\theta)^{16}$, thus, by Theorem 11,
$\frac{c^2[g'(m)]^2}{n} = \frac{81\theta^2}{n\theta^2}(\ln\theta)^{16} = \frac{81}{n}(\ln\theta)^{16}$
$Y_n \sim N\left([\ln(\theta)]^9, \frac{81}{n}(\ln\theta)^{16}\right)$

Q7. Consider a random sample from a Geometric distribution, $X_i \sim GEO(p)$. Let $W_i = e^{X_i}$ and $V_n = W_1 \times W_2 \times \cdots W_n$. $V_n^{\frac{1}{n}}$ converges in probability to a constant, identify the constant.

(5 marks)

Ans.

$E(\bar{X}_n) = \frac{1}{p}$, $V(\bar{X}_n) = \frac{1}{n}V(X) = \frac{1-p}{np^2}$
$P\left[|\overline{X_n} - \frac{1}{p}| \geq \epsilon\sqrt{\frac{1-p}{np^2}}\sqrt{\frac{np^2}{1-p}}\right] < \frac{(1-p)}{np^2\epsilon^2} \to 0$
$\therefore \bar{X}_n \xrightarrow{P} \frac{1}{p}$
$(V_n)^{1/n} = (W_1 \times W_2 \times \cdots \times W_n)^{1/n} = e^{\bar{X}_n}$
Thus, $(V_n)^{1/n} \xrightarrow{P} e^{\frac{1}{p}}$

Q8. Let $X_2, X_3, X_4, \ldots$ be a sequence of random variable such that

$$f_{X_n}(x) = \begin{cases} -n\ln\left(1 - \frac{1}{4n}\right)\left(1 - \frac{1}{4n}\right)^{nx} & x > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Find the limiting distribution of X_n.

(5 marks)

Ans.

$$\begin{aligned}
F_{X_n}(x) &= \int_0^x -n\ln\left(1-\tfrac{1}{4n}\right)\left(1-\tfrac{1}{4n}\right)^{nu} du \\
&= -n\ln\left(1-\tfrac{1}{4n}\right)\int_0^x e^{n\ln\left(1-\frac{1}{4n}\right)u} du \\
&= -n\ln\left(1-\tfrac{1}{4n}\right)\left[\frac{e^{n\ln\left(1-\frac{1}{4n}\right)u}}{n\ln\left(1-\frac{1}{4n}\right)}\right]_0^x \\
&= -\left[e^{n\ln\left(1-\frac{1}{4n}\right)x}-1\right] \\
&= 1-\left(1-\tfrac{1}{4n}\right)^{nx}
\end{aligned}$$

$$\begin{aligned}
\lim_{n\to\infty} F_{X_n}(x) &= \lim_{n\to\infty}\left(1-\left(1-\frac{1}{4n}\right)^{nx}\right) \\
&= 1-\lim_{n\to\infty}\left(1-\frac{1}{4n}\right)^{nx} \\
&= 1-e^{-x/4} \\
&= F_X(x)
\end{aligned}$$

$\Rightarrow X \sim Exp(4)$

Q9. Let $Y_n \sim GAM(\alpha n, \theta)$. Find the limiting distribution of $Z_n = \frac{Y_n - \alpha n\theta}{\sqrt{\alpha n}\theta}$ as $n \to \infty$, using moment generating function.

(10 marks)

Ans.

$M_{Y_n}(t) = (1-\theta t)^{-\alpha n}$

$$\begin{aligned}
M_{Z_n}(t) &= M_{\frac{Y_n-\alpha n\theta}{\sqrt{\alpha n}\theta}}(t) \\
&= e^{-\frac{\alpha n}{\sqrt{\alpha n}}t} M_{Y_n}\left(\tfrac{1}{\sqrt{\alpha n}\theta}t\right) \\
&= e^{-\frac{\alpha n}{\sqrt{\alpha n}}t}\left(1-\tfrac{1}{\sqrt{\alpha n}}t\right)^{-\alpha n} \\
&= e^{-\frac{\alpha n}{\sqrt{\alpha n}}t} e^{-\alpha n\ln\left(1-\frac{1}{\sqrt{\alpha n}}t\right)} \\
&= e^{-\frac{\alpha n}{\sqrt{\alpha n}}t} e^{-\alpha n\left[-\frac{1}{\sqrt{\alpha n}}t-\frac{1}{2\alpha n}t^2-\frac{1}{3\alpha n^{3/2}}t^3-\cdots\right]} \\
&= e^{-\frac{\sqrt{2\alpha n}}{2}t+\frac{\sqrt{2\alpha n}}{2}t+\frac{t^2}{2}+\frac{t^3}{3\sqrt{\alpha n}}-\cdots} \\
&= e^{\frac{t^2}{2}+\frac{t^3}{3\sqrt{\alpha n}}-\cdots}
\end{aligned}$$

$\lim_{\alpha n\to\infty} M_{\frac{Y_n-\alpha n}{\sqrt{2\alpha n}}}(t) = \lim_{n\to\infty} e^{\frac{t^2}{2}+\frac{t^3}{3\sqrt{\alpha n}}-\cdots} = e^{\frac{t^2}{2}}$

$\Rightarrow Z_n = \frac{Y_n-\alpha n\theta}{\sqrt{\alpha n}\theta} \xrightarrow{d} N(0,1)$

Q10. Suppose that $X_i \sim N(\mu,\sigma^2), i = 1,\ldots,14$, $Z_j \sim N(0,1), j = 1,\ldots,9$, and $W_k \sim \chi^2(50), k = 1,\ldots,13$ and all random variables are independent. State

the distribution of each of the following variables if it is a "named" distribution. [For example $X_1 + X_2 \sim N(2\mu, 2\sigma^2)$]

(a) $\frac{8\sum_{i=1}^{14}(X_i-\bar{X})^2}{13\sigma^2\sum_{j=1}^{9}(Z_j-\bar{Z})^2}$.

(b) $\frac{8\sum_{k=1}^{9}W_k}{450\sum_{j=1}^{9}(Z_j-\bar{Z})^2}$

(c) $\frac{\sqrt{700}(\bar{X}-\mu)}{\sigma\sqrt{W_1}}$

(d) $\frac{W_1}{W_1+W_2+W_3+W_4}$

(e) $\frac{Z_i^2/W_1}{1+Z_1^2/W1}$

(f) $\frac{\frac{\sum_{k=1}^{9}W_k}{\sum_{j=1}^{9}(Z_j-\bar{Z})^2}}{1+\frac{\sum_{k=1}^{9}W_k}{\sum_{j=1}^{9}(Z_j-\bar{Z})^2}}$

(20 marks)

Ans.

(a) $\frac{\sum_{i=1}^{14}(X_i-\bar{X})^2}{\sigma^2} \sim \chi^2(13)$

$\sum_{j=1}^{9}(Z_j-\bar{Z})^2 \sim \chi^2(8)$

$\frac{\frac{\sum_{i=1}^{14}(X_i-\bar{X})^2}{13\sigma^2}}{\sum_{j=1}^{9}(Z_j-\bar{Z})^2/8} \sim F(13,8)$

Thus, $\boxed{\frac{8\sum_{i=1}^{14}(X_i-\bar{X})^2}{13\sigma^2\sum_{j=1}^{9}(Z_j-\bar{Z})^2} \sim F(13,8)}$.

(b) $\sum_{k=1}^{9}W_k \sim \chi^2(450)$

$\sum_{j=1}^{9}(z_j-\bar{Z})^2 \sim \chi^2(8)$

$\frac{\sum_{k=1}^{9}W_k/450}{\sum_{j=1}^{9}(z_j-\bar{Z})^2/8} \sim F(450,8)$

Thus, $\boxed{\frac{8\sum_{k=1}^{9}W_k}{450\sum_{j=1}^{9}(z_j-\bar{Z})^2} \sim F(450,8)}$

(c) $\frac{\frac{\sqrt{14}(\bar{X}-\mu)}{\sigma}}{\sqrt{W_1/50}} \sim N(0,1)$

$\frac{\frac{\sqrt{14}(\bar{X}-\mu)}{\sigma}}{\sqrt{W_1/50}} \sim t(50)$

Thus, $\boxed{\frac{\sqrt{700}(\bar{X}-\mu)}{\sigma\sqrt{W_1}} \sim t(50)}$

(d) $\frac{W_1}{W_1+W_2+W3+W_4} = \frac{W_1}{W_1+(W_2+W3+W_4)}$

$W_1 \sim \chi^2(50) \sim GAM(25.0,2)$, $W_2+W_3+W_4 \sim \chi(150) \sim GAM(75.0,2)$

Thus, $\frac{W_1}{W_1+W_2+W3+W_4} \sim Beta(25.0,75.0)$

(e) $Z_1^2 \sim \chi^2(1)$

$\frac{Z_1^2}{W_1/50} \sim F(1,50)$

$\frac{(\frac{1}{50})\frac{Z_1^2}{W_1/50}}{1+(\frac{1}{50})\frac{Z_1^2}{W_1/50}} = \frac{Z_i^2/W_1}{1+Z_1^2/W1} \sim Beta(1/2,25.0)$

(f) $\sum_{k=1}^{9} W_k \sim (450)$

$\sum_{j=1}^{9}(z_j - \bar{Z})^2 \sim \chi^2(8)$

$\frac{\sum_{k=1}^{9} W_k/450}{\sum_{j=1}^{9}(z_j-\bar{Z})^2/8} \sim F(450, 8)$

Thus, $\frac{8\sum_{k=1}^{9} W_k}{450\sum_{j=1}^{9}(z_j-\bar{Z})^2} \sim F(450, 8)$

$$\frac{(\frac{450}{8})\frac{8\sum_{k=1}^{9} W_k}{450\sum_{j=1}^{9}(Z_j-\bar{Z})^2}}{1+(\frac{450}{8})\frac{8\sum_{k=1}^{9} W_k}{450\sum_{j=1}^{9}(Z_j-\bar{Z})^2}} = \frac{\frac{\sum_{k=1}^{9} W_k}{\sum_{j=1}^{9}(Z_j-\bar{Z})^2}}{1+\frac{\sum_{k=1}^{9} W_k}{\sum_{j=1}^{9}(Z_j-\bar{Z})^2}} \sim Beta(225.0, 4.0)$$